

Memo to the Emergency Task Force:

Will Humankind Survive the Zombie Apocalypse?

Introduction: The Emergency Task Force is getting reports of cases of a new kind of disease that turns humans into zombies. It appears to be transmitted via bites. The disease is now ranked as an international health emergency and we urgently need information from models to allocate our resources to maximum effect. At the moment, two major mitigation strategies are discussed: developing a vaccine that reduces human susceptibility to the zombie-disease or arming and training large parts of the population in self-defence techniques to improve their chances of fighting off zombie attackers. However, we only have the resources to pursue one of these strategies.

Technical Details: Experts agree that the model published in the book by Griffiths and Higham¹ (see Reference section in StudIP) is the most faithful representation of the dynamics of Zombie disease. Current data puts the parameters of the model at $\alpha = 0.5$, $\beta = 0.6$, although uncertainties are large and it cannot be ruled out that these estimates are up $\pm 25\%$ off from the true values. At the moment, no cases of zombies resurrecting have been observed and therefore we assume $\zeta = 0$.

Task: Your task is to inform our decision making by writing a memo (not more than two pages long) that

1. Provides a concise discussion of the underlying assumptions and resulting limitations of the model. Describe how you represent the two interventions in the model.
2. Give example solutions highlighting all possible scenarios within the plausible range of parameters for an example population of 1000 humans over the course of one year.
3. Give a clear recommendation for one strategy based on evidence: train and arm humans to increase their capability to fight off zombies by roughly 50% or develop a vaccine that reduces the chance of a zombie infecting a human by about half.

Generate results by using a suitable ODE solver in Python.

Submission: Your submission consists of two parts

1. A technical report of up to 2 pages as PDF submitted to ILIAS, please see StudIP for the link. Your file name shall be `LASTNAME_GIVENNAME.pdf`
2. A Python code verified in YAPS. The online examination system YAPS will be open from **27.04. to 05.05.**, see the file `YAPS_how_to_use.pdf` in StudIP/ILIAS for the link and further information.

Your code must implement a Python function `zombie(strategy, pa, pb)` that returns a time grid vector $\mathbf{t}$ of length $n = 101$ for the simulation period of one year, i.e., $\mathbf{t}[0] = 0$ and $\mathbf{t}[100] = 1$, and a corresponding matrix $\mathbf{x}$ with n rows and three columns such that

$\mathbf{x}[i,0]$ is the number of humans
 $\mathbf{x}[i,1]$ is the number of zombies
 $\mathbf{x}[i,2]$ is the number of removed zombies

at time $\mathbf{t}[i]$ for $i = 0, \dots, n - 1$. The parameter `strategy` stands for a mitigation strategy to encounter the zombie apocalypse. The following values are allowed:

0 : no intervention
 1 : training and arming humans
 2 : vaccination of humans

The parameters `pa`, `pb` represent the respective uncertainties in the model parameters α (efficiency of humans killing zombies) and β (efficiency of zombies infecting humans).

Generate the results by using a suitable ODE solver in Python.

¹David F. GriffithsDesmond J. Higham, *Numerical Methods for Ordinary Differential Equations*, Springer-Verlag London Limited 2010

A sketch for the function `zombie(...)` that you can complete is given below.

```
import numpy as np
from scipy.integrate import odeint

def zombie(strategy,pa,pb):
    """
    zombie(strategy,pa,pb)

    This function computes the evolution of the populations of humans and zombies
    according to a chosen mitigation strategy.

    Parameters
    -----
    strategy : integer
        mitigation strategy
        0 : no intervention
        1 : training and arming humans
        2 : vaccination of humans
    pa : real number
        uncertainty in the parameter alpha (= efficiency of humans killing zombies)
    pb : real number
        uncertainty in the parameter beta (= efficiency of zombies infecting humans)

    Returns
    -----
    t : numpy.array
        vector containing the time grid of the integration period of 1 year, t = [0,...,1]
    x : numpy.array
        x[i,0] : number of humans at time t[i]
        x[i,1] : number of zombies at time t[i]
        x[i,2] : number of removed zombies at time t[i]

    Example
    -----
    >>> t, x = zombie(0,0,0)          # Baseline scenario without intervention and
                                     # with given values of alpha and beta
    >>> t, x = zombie(1,-0.25,0.25)   # Strategy 1 in worst case scenario where alpha is
                                     # 25% decreased and beta is 25% increased.
    """

    #####
    #edit the code here#
    #####

    return ##
```

Before you submit your implementation of the function `zombie` in YAPS you can a priori check it offline with the main program `__main__.py` which is available in StudIP/ILIAS. (Finally, YAPS will run additional tests of your code which we do not disclose.)

Deadline for submission is Tuesday 5 May 2026 at midnight.

Dr. wich. tig. N. Otreál
Chair of the Emergency Task Force

Report structure. Please structure your report using the following sections.

1. Introduction: this should briefly describe the problem, introduce the model, explain the question to be answered and describe how the interventions are modeled.
2. Scenarios: present example scenarios that illustrate the possible outcomes you identified.
3. Recommendation: provide your recommendation and reference your scenarios to provide evidence.

Please use 10pt or 11pt fontsize and a standard font like Arial, Helvetica or something similar. You must respect the limit of two pages, anything beyond two pages will not be considered for marking.

Some general tips for writing reports.

- Do not mix answers to different questions. Every question deserves its own answer that is clearly separate from the others. Do not let the reader figure out which question you are addressing in a given part of the text.
- Every figure needs to be referenced and discussed in the text. Each figure should have a clear “message” associated with it which should be clearly spelled out in the text. If you can’t say what the message of a figure is, you either need to think more about it before writing it down or the figure doesn’t really have one - remove it, then.

Resist the urge to pad your report with many figures that do not contribute to the key points its making. It will not improve its quality and just look cluttered.

- Keep sentences short. At least one round of proof-reading should be dedicated to identify overly long sentences and to breaking them down.
- At least one additional round of proof-reading should be used to identify and delete waffle. Say what you want and need to say, not more, not less. Being concise is a virtue.
- Use paragraphs.
- Make sure figures are readable, also when printed in black and white. Whenever possible, use vector formats like PDF for graphs as they scale without loss.